

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Solid wastes - Set 1

Mark scheme

AQA spec 3.4.3.2.12; Pollution; textbook pp. 279-285

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 2 points below. 1 Landfill fell from 79% to 14% (large decrease) 1 Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest) Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Incineration → burning waste, usually with energy recovery • recycling → reprocessing a material to make something new Do not credit • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Question	Marking guidance	Marks
01.2	Any 3 of 5 1 mark each for any 3 of the 5 points below, in any order, maximum 3. 1 Organic matter decays anaerobically releasing methane (and CO₂) - greenhouse gases 1 Toxic leachate may leak into groundwater / rivers from poorly managed sites 1 Uses large areas of land – loss of habitats / farmland – contamination prevents later use 1 Resource value of metals, glass, plastics is lost 1 Transport noise / congestion / litter / pests / odour Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • leachate → polluted liquid draining out of waste Do not credit • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
01.3	Any 4 of 7 1 mark each for any 4 of the 7 points below, in any order, maximum 4. 1 Polymer / clay liner to prevent escape of leachate 1 Collection and treatment of leachate 1 Collection of methane and use as a fuel 1 Impermeable cap (clay and polymer) when complete to stop water entering 1 Regular covering with soil (pests, odour) – deodorising sprays 1 Perimeter fence to trap wind-blown litter 1 Separation and recording of waste types – reactive / flammable wastes not dumped together Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • leachate → polluted liquid draining out of waste • Impermeable → does not let water through Do not credit • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	4
01.4	1 mark for the point below. 1 Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • incineration → burning waste, usually with energy recovery • recycling → reprocessing a material to make something new Do not credit • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	
02.1	Any 3 of 5 1 mark each for any 3 of the 5 points below, in any order, maximum 3. 1 Volume of ash is much less than the original waste (less landfill) 1 Heat used for district heating / electricity generation (energy recovery) 1 No sorting / complicated management needed 1 Destroys pathogens / flammable organic wastes converted to CO₂ and water 1 Refuse-derived fuel can replace fossil fuels (eg cement kilns) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
02.2	1 mark each 1 mark for each of the 4 points below. 1 High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) 1 Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone OR removes heavy metals for storage at high pH 1 Wet waste needs extra fuel to burn and lowers combustion temperature – composting recovers its nutrient value 1 Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Recycling → reprocessing a material to make something new Do not credit 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
02.3	1 mark each 1 mark for each of the 2 points below. 1 Ash may contain concentrated heavy metals / toxic residues 1 Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Question	Marking guidance	Marks
02.4	Any 1 of 2 1 mark each for any 1 of the 2 points below, in any order, maximum 1. 1 Fuel (natural gas) needed to burn wet / non-flammable wastes 1 Cost of treating effluent gases (SO₂, NO_x, HCl, smoke) Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. incineration → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
03.1	2 marks per point 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. 2 Asbestos: fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma – disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) 2 Cyanide: very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested – disposal: incineration so carbon and nitrogen are separated and oxidised to CO₂ and NO_x (much smaller problem) 2 High-level radioactive waste: intense ionising radiation, heat generation, long half-lives – disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. incineration → burning waste, usually with energy recovery ionising radiation → radiation with enough energy to remove electrons and form ions vitrification → sealing waste into glass Do not credit 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
03.2	1 mark for the point below. 1 The fibres are bound / cannot become airborne so cannot be inhaled Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.3	1 mark each 1 mark for each of the 2 points below. 1 Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material 1 The solid cement immobilises the waste so it cannot leak / leach out Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. impermeable → does not let water through Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Question	Marking guidance	Marks
03.4	1 mark for the point below. 1 Even if the glass shatters the waste stays encapsulated in the fragments – glass is more stable / resistant to heat and leaching over very long timescales Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. vitrification → sealing waste into glass leaching → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	
04.1	1 mark each 1 mark for each of the 6 points below. 1 Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes 1 Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH 1 Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow 1 Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser 1 Unattractive topography - landscaping / reshaping 1 Dust / heavy metals - cover and keep at high pH Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. leachate → polluted liquid draining out of waste Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	6
04.2	1 mark each 1 mark for each of the 2 points below. 1 So that plant roots cannot reach the toxic layer 1 Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	1 mark each 1 mark for each of the 2 points below. 1 Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions 1 Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
05.1	1 mark each 1 mark for each of the 4 points below. 1 Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them 1 Built-in obsolescence: products designed to have short lives / wear out / go out of fashion so customers upgrade 1 Disposable products (razors, lighters, pens) that must be replaced 1 Over-packaging to make products attractive OR increases wastes from manufacture and mineral extraction Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	4
05.2	Any 3 of 7 1 mark each for any 3 of the 7 points below, in any order, maximum 3. 1 Properties of the waste: toxicity, flammability, degradability, radioactivity 1 Population density (collection costs) 1 Land availability for landfill 1 Availability of recycling technology / markets 1 Legislation / regulatory framework (landfill tax) 1 Public willingness to recycle / environmental awareness 1 Processing costs / household income Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. recycling → reprocessing a material to make something new Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
05.3	1 mark each 1 mark for each of the 2 points below. 1 Advantage: aerobic decomposition produces little methane – nutrients / organic matter returned to soil (circular economy) – reduces landfill volume 1 Disadvantage: needs separation / collection of clean organic waste – odour / pests / pathogens if poorly managed – low-value product – not suitable for mixed waste Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Question	Marking guidance	Marks
05.4	1 mark for the point below. 1 Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
06.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 A: Aerobic decomposition produces little methane and returns nutrients to the soil	1
06.2	1 mark each 1 mark for each of the 2 points below. 1 It may contain concentrated heavy metals / toxic residues 1 It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
06.3	Any 2 of 4 1 mark each for any 2 of the 4 points below, in any order, maximum 2. 1 Impermeable polymer / clay liner to stop leachate escaping 1 Collection and treatment of leachate 1 Collection of methane and use as fuel 1 Daily covering of waste / final capping and restoration Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Impermeable → does not let water through leachate → polluted liquid draining out of waste Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	